ASSIGNMENT : 02

LINEAR ALGEBRA

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"DETERMINANT & ITS PROPERTIES"

- **What is a determinant?**

Determinant is a scalar value which is the function of entries of a square matrix. It characterizes properties of matrix.

Determinant of 2x2 matrix:

Determinant of a 2x2 matrix is calculated as:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \Longrightarrow \begin{vmatrix} a & b \\ c & d \end{vmatrix} = (a)(d) - (b)(c)$$

Determinant of 3x3 matrix:

Determinant of a 3x3 matrix is calculated as:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \Longrightarrow \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

$$= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$= a\left((e)(i) - (f)(h)\right) - b\left((d)(i) - (f)(g)\right) + c\left((d)(h) - (e)(g)\right)$$

Properties of determinants:

1. Reflection Property:

The determinant remains unchanged if rows are changed into columns & columns into rows i.e. transpose of a matrix is taken.

Example:

$$\begin{vmatrix} 1 & 4 & -2 \\ 3 & 6 & 4 \\ 0 & 7 & 5 \end{vmatrix} = (1)(30-28) - (4)(15-0) + (-2)(21-0)$$

$$= 2 - 60 - 42 = -100$$

→ Taking transpose of matrix.

$$\begin{vmatrix} 1 & 3 & 0 \\ 4 & 6 & 7 \\ -2 & 4 & 5 \end{vmatrix} = (1)(30-28) - (3)(20+14) + 0$$

$$= 2 - 102 = -100 \quad \text{(proved)}$$

2. All-zero Property:

If every element of a row or column of a matrix is zero, than the determinant is zero.

Example:

$$\begin{vmatrix} 0 & 2 & 4 \\ 0 & -1 & 5 \\ 0 & 7 & 3 \end{vmatrix} = (0) - 2(0-0) + 4(0-0) = 0 \quad \text{(proved)}$$

3. Proportionality property:

If all the elements of a row or column are identical to all elements of some other row or column, then the determinant is zero.

Example:

$$\begin{vmatrix} 4 & -2 & 4 \\ 6 & 4 & 6 \\ 7 & 5 & 7 \end{vmatrix} = 4(28-30) + 2(42-42) + 4(30-28)$$

$= -8 + 0 + 8 = 0.$ (proved).

4. **Switching Property:**

If we interchange any two rows or columns of a matrix, than the determinant of new matrix changes its sign.

Example:

$$\begin{vmatrix} 1 & 4 & -2 \\ 3 & 6 & 4 \\ 0 & 7 & 5 \end{vmatrix} = 1(30-28) - 4(15-0) + (-2)(21-0)$$

$= 20 - 60 - 42 = -100.$

→ Interchanging R1 & R3.

$$\begin{vmatrix} 0 & 7 & 5 \\ 3 & 6 & 4 \\ 1 & 4 & -2 \end{vmatrix} = 0 - 7(-6-4) + 5(12-6) = 70 + 30 = 100$$

(proved)

5. **Scalar Multiple Property:**

If every element of a row or a column is multiplied by any non-zero constant, then the determinant gets multiplied by the same constant.

Example:

$$\begin{vmatrix} 1 & 4 & -2 \\ 3 & 6 & 4 \\ 0 & 7 & 5 \end{vmatrix} = -100 \xrightarrow{R_1 \times 2} \begin{vmatrix} 2 & 8 & -4 \\ 3 & 6 & 4 \\ 0 & 7 & 5 \end{vmatrix}$$

$= 2(30-28) - 8(15-0) - 4(21-0)$

$= 4 - 120 - 84 = -200$ (proved).

6. Sum Property:

If elements of a row or a column are expressed as sum of two or more terms, then the determinant can be expressed as sum of two or more determinants.

Example:

$$\begin{vmatrix} 2+1 & 1 & 2 \\ 3+(-2) & 0 & 5 \\ 4+2 & 3 & 7 \end{vmatrix} = 3(0-15) - 1(7-30) + 2(3-0)$$

$$= -45 + 23 + 6 = -16.$$

$$= \begin{vmatrix} 2 & 1 & 2 \\ 3 & 0 & 5 \\ 4 & 3 & 7 \end{vmatrix} + \begin{vmatrix} 1 & 1 & 2 \\ -2 & 0 & 5 \\ 2 & 3 & 7 \end{vmatrix}$$

$$= [2(-15) - 1(21-20) + 2(9)] + [1(15) - 1(-14-10) + 2(6)]$$

$$= [-30 - 1 + 18] + [-15 + 24 - 12]$$

$$= -13 - 3 = -16. \text{ (proved).}$$

7. Property of Invariance:

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 + \alpha b_1 + \beta c_1 & b_1 & c_1 \\ a_2 + \alpha b_2 & \beta c_2 & b_2 & c_2 \\ a_3 + \alpha b_3 & \beta c_3 & b_3 & c_3 \end{vmatrix}$$

Example:

$$\begin{vmatrix} 1 & 1 & 2 \\ -2 & 0 & 5 \\ 2 & 3 & 7 \end{vmatrix} = 1(15) - 1(-14-10) + 2(6) = -3.$$

$$\begin{vmatrix} 1+1+2 & 1 & 2 \\ -2+0+5 & 0 & 5 \\ 2+3+7 & 3 & 7 \end{vmatrix} = \begin{vmatrix} 4 & 1 & 2 \\ 3 & 0 & 5 \\ 2 & 3 & 7 \end{vmatrix}$$

$$= 4(-15) - 1(21-60) + 2(9)$$

$$= -60 + 39 + 18 = -3. \text{ (proved).}$$

8. Factor Property:

If determinant becomes zero when we put $x = \alpha$ then $(x - \alpha)$ is a factor of Δ.

Example:

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \Rightarrow \Delta_1 = \begin{vmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{vmatrix}$$

where C_{ij} is the co-factor of elements a_{ij} in Δ.

9. Triangle Property:

If all the elements of a determinant above -or below the main diagonal consist of zeros, then the determinant is equal to the product of diagonal elements.

Example:

$$\begin{vmatrix} 2 & 0 & 0 \\ 4 & 3 & 0 \\ 1 & 5 & 1 \end{vmatrix} = 2(3) - 0 - 0 = 6$$

or $(2)(3)(1) = 6$ (proved).